

Exploratory Data Analysis (EDA) Report

Project: Social Engine Analytics Platform | Data Vortex - Round 1

Author: Himanshu Gaur | Course: Computer Science & Software Engineering

1. Executive Summary

This report outlines the rigorous exploratory data analysis (EDA) and robust preprocessing pipeline executed for the Social Engine analytics datasets (Social_Engine_Users and Social_Engine_Posts_Corrupted). The primary objective of this phase was to transform raw, uncleaned, and corrupted data into a pristine, production-ready state tailored for downstream machine learning models and Natural Language Processing (NLP) tasks.

By implementing advanced statistical imputation, custom mixed-format timestamp parsing, and strict duplicate elimination, the dataset maintains high fidelity while preventing common predictive model biases such as zero-inflation.

2. Dataset Overview & Architecture

The project architecture integrates two core relational datasets structured for social media interaction tracking:

- **Users Dataset (Social_Engine_Users_Cleaned.csv):** Captures user profile metadata, account registration dates, and aggregate follower metrics across global user tiers.
- **Posts Dataset (Social_Engine_Posts_Cleaned.csv):** Captures post-level engagement metrics (likes, shares, comments), textual content, platform origin sources, and publication timestamps.

Key Preprocessing Steps Performed:

- 1. Duplicate Elimination:** Redundant duplicate records were purged from both tables to eliminate skewness in frequency and demographic distributions.
- 2. Advanced Missing Value Imputation:** Rather than applying crude zero-filling (which artificially distorts feature distributions), missing numerical values in likes, shares, and comments were imputed using their statistical medians. Categorical missing platforms were labeled as 'Unknown', and text contents were left blank ("") to ensure optimal NLP tokenization.
- 3. Robust Timestamp Normalization:** A custom robust parser was engineered to seamlessly unify mixed date formats—including Unix epoch integers, ISO strings, and standard date formats—without throwing warnings or corrupting temporal points.

3. Exploratory Data Analysis & Statistical Findings

A. User Follower Demographics

An analysis of user follower counts reveals a distinct right-skewed distribution, where the median follower count is substantially lower than the mean due to power-law user concentrations. The average user account holds approximately 24,964 followers, with primary activity clusters located in high-density metropolitan regions.

B. Platform Engagement Analysis

Cumulative cross-platform activity tracking across Facebook, YouTube, Instagram, Reddit, and Twitter demonstrates that video-centric and legacy social platforms (specifically YouTube and Facebook) capture the highest aggregate interaction volume in terms of raw likes. However, per-post engagement medians remain remarkably stable across secondary networks, reflecting healthy niche community participation.

C. Interaction Correlation Insights

A correlation matrix heatmap evaluating likes, shares, and comments indicates near-zero linear correlation coefficients among the metrics. This statistical insight proves that user engagement behaviors operate independently; a post receiving high likes does not automatically or proportionally trigger higher shares or comments, highlighting distinct user interaction intents.

D. Summary Statistics Table

Dataset Feature	Imputation / Cleaning Method	Impact on Downstream ML/NLP
Numerical Metrics (likes, shares, comments)	Median Imputation	Prevents zero-inflation bias and skewness in ML models.
Text Content (text_content)	Preserved Blank (Empty Strings)	Ensures clean tokenization and embedding in NLP models.
Timestamps (timestamp)	Robust Custom Parser (Epoch/ISO)	Enables flawless time-series analysis without parsing errors.

4. Conclusion & Next Steps

The exploratory data analysis and preprocessing phase is fully complete, warning-free, and exported into production-ready CSV files (Social_Engine_Users_Cleaned.csv and Social_Engine_Posts_Cleaned.csv). All code notebooks, visualization charts (platform_engagement.png, follower_distribution.png, engagement_correlation.png), and documentation have been version-controlled and structured into a GitHub repository, establishing a robust, reproducible workflow ready for Round 2 predictive modeling.